Transcriptomic Profiling of Drug Response Reveals Dysregulated Pathways in Inflammatory and Apoptotic Signaling

Authors: C. Pharmacologist, D. Bioinformatician

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Abstract

Background: Understanding the transcriptomic response to pharmacological intervention is critical for drug development. This study profiled 20 genes in a cell line model treated with Compound X, a novel anti-inflammatory agent, to characterize pathway-level changes.

Methods: RNA-seq count data for 20 genes were profiled across 6 samples (3 Control, 3 Drug-treated). Differential expression analysis was performed using DESeq2 v1.42.0 in R v4.3.0. Gene Ontology enrichment analysis was performed using clusterProfiler. KEGG pathway visualization was performed using pathview.

Results: Four genes were significantly upregulated: IL6 (log2FC = 3.0, padj = 0.0003), TNF (log2FC = 2.5, padj = 0.001), CXCL8 (log2FC = 2.2, padj = 0.002), and IL1B (log2FC = 2.0, padj = 0.005). Two genes were significantly downregulated: BCL2 (log2FC = -2.2, padj = 0.0008) and MCL1 (log2FC = -1.8, padj = 0.003). GO enrichment of the significant genes revealed enrichment for inflammatory response and cytokine signaling pathways. KEGG pathway visualization confirmed activation of the TNF signaling pathway.

Conclusions: Compound X induces a pronounced inflammatory transcriptional program while suppressing anti-apoptotic gene expression. The dysregulation of TNF and NF-kappa B signaling pathways suggests a complex mechanism of action involving both pro-inflammatory and pro-apoptotic effects.

Introduction

RNA-sequencing enables systematic characterization of drug-induced transcriptomic changes. While differential expression analysis identifies individual gene-level changes, pathway enrichment analysis provides biological context by revealing coordinated changes in functional gene sets. The integration of differential expression, gene ontology enrichment, and pathway visualization constitutes a standard multi-step analysis pipeline in transcriptomic pharmacology.

This study examines the transcriptomic response of 20 genes to Compound X treatment. We employ a three-stage analysis: (1) differential expression analysis with DESeq2, (2) Gene Ontology enrichment with clusterProfiler, and (3) KEGG pathway visualization with pathview. The 20-gene panel includes key inflammatory cytokines, apoptotic regulators, and housekeeping controls, enabling a focused but biologically informative analysis.

Methods

Sample Preparation

RNA-seq was performed on 6 samples: - **Control group:** 3 biological replicates (Control_1, Control_2, Control_3) - **Drug-treated group:** 3 biological replicates (Drug_1, Drug_2, Drug_3)

Each sample was profiled for the same 20 genes. The count matrix is provided as `counts.csv` (Supplementary Table S1).

Differential Expression Analysis

Count data were normalized and analyzed using DESeq2 v1.42.0 in R v4.3.0. The analysis followed the standard DESeq2 workflow:

1. Load count matrix from CSV
2. Create DESeqDataSet with design formula `~ condition`
3. Run DESeq2 with default parameters
4. Apply log2 fold change shrinkage with `lfcShrink(type = "apeglm")`
5. Extract results at adjusted p-value < 0.05

GO Enrichment Analysis

Gene Ontology enrichment analysis was performed on the set of differentially expressed genes using clusterProfiler. Enriched Biological Process terms were visualized as a bar plot.

KEGG Pathway Visualization

KEGG pathway visualization was performed using pathview to map differentially expressed genes onto the TNF signaling pathway.

Visualization

Figure 1: Volcano plot was generated using ggplot2 v3.5.0, displaying -log10(adjusted p-value) on the y-axis against log2 fold change on the x-axis. Significant genes (padj < 0.05) were highlighted in red. Genes with mean normalized count < 10 across all samples were excluded prior to differential expression analysis.

Figure 2: GO enrichment bar plot showing the top 10 enriched Biological Process terms.

Figure 3: KEGG pathway diagram of the TNF signaling pathway, with upregulated genes colored red and downregulated genes colored blue.

Software Versions

Software	Version	Purpose
R	4.3.0	Statistical computing environment
DESeq2	1.42.0	Differential expression analysis
ggplot2	3.5.0	Data visualization
apeglm	1.24.0	Log2 fold change shrinkage
clusterProfiler	—	GO enrichment analysis
pathview	—	KEGG pathway visualization

Results

Differential Expression

DESeq2 identified 6 significantly differentially expressed genes (padj < 0.05):

Table 1: Differential Expression Results

Gene	log2FC	padj	Direction
IL6	3.0	0.0003	Upregulated
TNF	2.5	0.001	Upregulated
CXCL8	2.2	0.002	Upregulated
IL1B	2.0	0.005	Upregulated
BCL2	-2.2	0.0008	Downregulated
MCL1	-1.8	0.003	Downregulated

The complete differential expression results for all 20 genes are available in Supplementary Table S2.

GO Enrichment

Gene Ontology enrichment analysis of the 6 significant genes revealed enrichment for pathways related to inflammatory response, cytokine signaling, and apoptotic regulation.

Table 2: Top 5 GO Biological Process Terms

GO Term	Description	GeneRatio	p.adjust
GO:0006954	inflammatory response	4/6	0.0002

GO Term	Description	GeneRatio	p.adjust
GO:0019221	cytokine-mediated signaling pathway	4/6	0.0005
GO:0071345	cellular response to cytokine stimulus	3/6	0.001
GO:0043067	regulation of programmed cell death	3/6	0.003
GO:0006915	apoptotic process	3/6	0.005

Significance threshold: q-value < 0.05 (Benjamini-Hochberg adjusted)

Figure 1: Volcano Plot

Figure 1 is a volcano plot showing -log10(adjusted p-value) vs log2 fold change for all genes passing the expression filter. The four upregulated genes (IL6, TNF, CXCL8, IL1B) appear in the upper-right quadrant. The two downregulated genes (BCL2, MCL1) appear in the upper-left quadrant. The remaining 14 genes cluster near the origin.

Figure 2: GO Enrichment Bar Plot

Figure 2 is a bar plot of the top 10 enriched GO Biological Process terms, ordered by -log10(q-value). Terms related to inflammatory response and cytokine signaling show the strongest enrichment.

Figure 3: KEGG Pathway Diagram

Figure 3 is a KEGG pathway diagram of the TNF signaling pathway (hsa04668). Upregulated genes (IL6, TNF, CXCL8, IL1B) are shown in red, and downregulated genes (BCL2, MCL1) are shown in blue, confirming pathway-level dysregulation consistent with the GO enrichment results.

Data Availability

The count matrix is available as Supplementary Table S1 (`counts.csv`), containing 20 rows (genes) and 6 columns (samples). The dataset is also deposited at GEO under accession **GSE99999**.

The complete differential expression results are provided in Supplementary Table S2.

The analysis script is available at `https://github.com/example/drug-response-analysis`.

References

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